

Regression Analysis: NSS 69TH Round (Drinking Water, Sanitation, Hygiene And Housing Conditions)



By: Anjali, Maharshi

CONSUMER EXPENDITURE ~ SOCIAL GROUPS

MOTIVATION

By doing this regression we wanted to understand the effect of belonging to different social groups on the patterns of consumer expenditure. We are aware that structurally there are inequalities between different social groups cultivated by certain practices. This paper assumes that these inequalities will affect their economic conditions and thereby their consumer patterns. This regression is being carried out in order to understand the type and extent of the effect that these conditions will have on household's consumption expenditure.

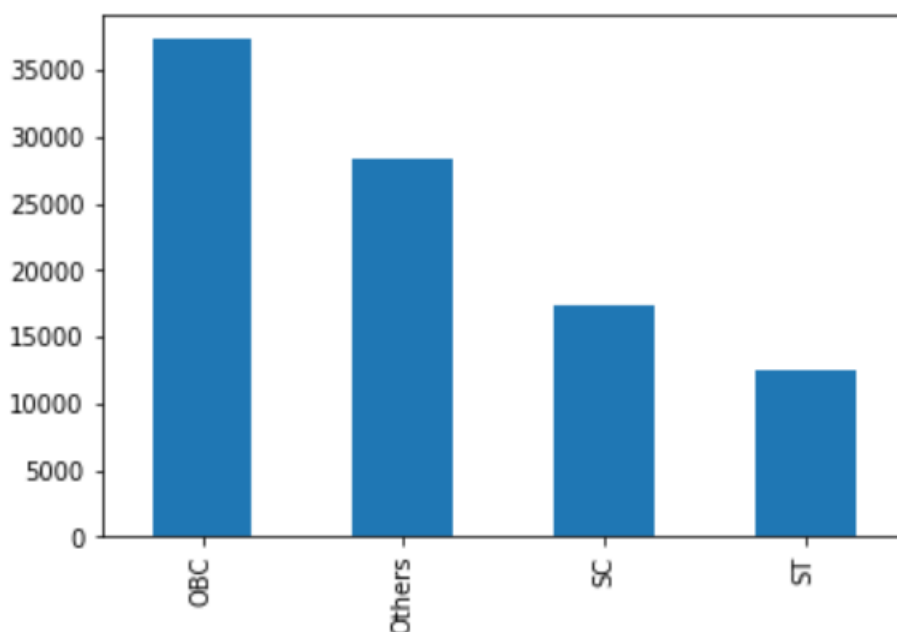
METHODOLOGY

We started off by loading our dataset and merging the required blocks/sub-blocks together as the data was divided as such. Following this, we created a dataset with all the required columns addressing the questions we were investigating. We dropped any entries which had NA values in the dataset and proceeded to create dummy variables representing the different component of the individual's social identity that we were interested in.

We then ran multiple regressions 1(regc)-including only the caste divisions as the independent variables, 2(regr)-including caste+religion divisions as the independent variables, 3(rega)-including caste+religion+age divisions as the independent variables and finally 4(reg)-which includes caste+religion+age+rural divisions as the independent variables. The dependent variable stayed constant across the different regressions, we set it as $\ln(\text{consumption expenditure})$. We chose to take the log values of the consumption expenditure to correct for the random spread of the absolute values given in the dataset. Upon generating our desired regressions we summarised the results and exported them as a table(Table-1).

REG C: $\ln(\text{Consumption Expenditure}) \sim \text{Caste}$

Distribution of households based on Caste

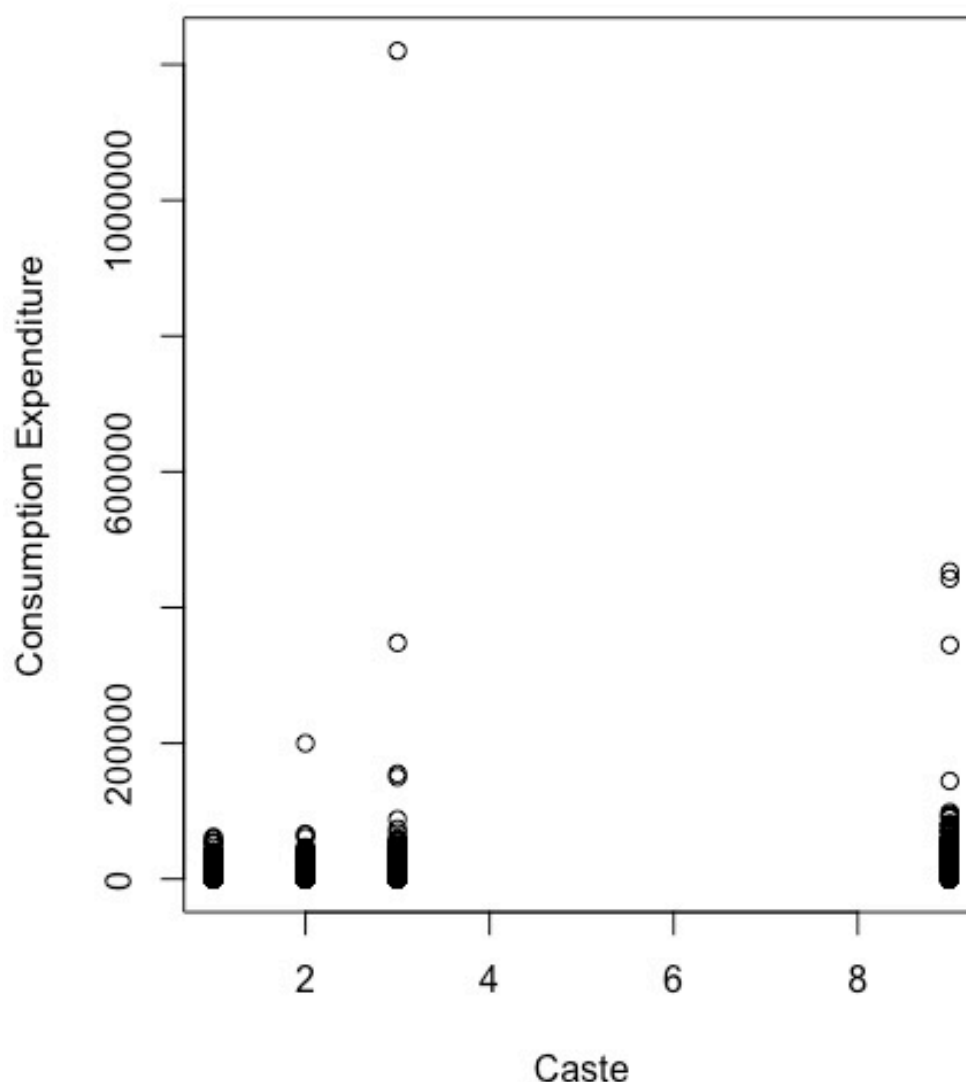


Graph 1

Graph 1 visualises the spread of the households among the different caste categories, This helps us understand the distribution of the data better when studying the regression.

Like mentioned, REGC aims to understand the relationship between the $\ln(\text{household consumption expenditure})$ and the Castes that they belong to. Graph 2 is a scatter plot that visualises the relationship between the variables. It reflects the spread of the 4 caste categories that we interact with through the dataset. The codes associated with it are used as the x-axis limits, where 1 = STs, 2= SCs, 3= OBSc and 9= Other castes. The spread is visualised in accordance and shows us the expenditure distribution among these groups.

Relationship between Caste and Consumption Expenditure



Graph 2

X-Axis Key - 1: ST, 2: SC, 3: OBC, 9: Others

TESTING THE SUITABILITY OF THE REGRESSION (REGC)

- Linearity

A plot with a best line between social group(caste) and the $\ln(\text{consumption expenditure})$ was made(similar to Graph2) to see the nature of the relationship between the variables. In this case it was found that there is a linear relationship where the best fit line is very slightly sloping upwards. This indicates that consumption expenditure slightly increases in percentage terms across the respective groups (SC to 'other' castes). This supports our hypothesis and speaks to the accuracy of the regression.

- Correlation coefficient

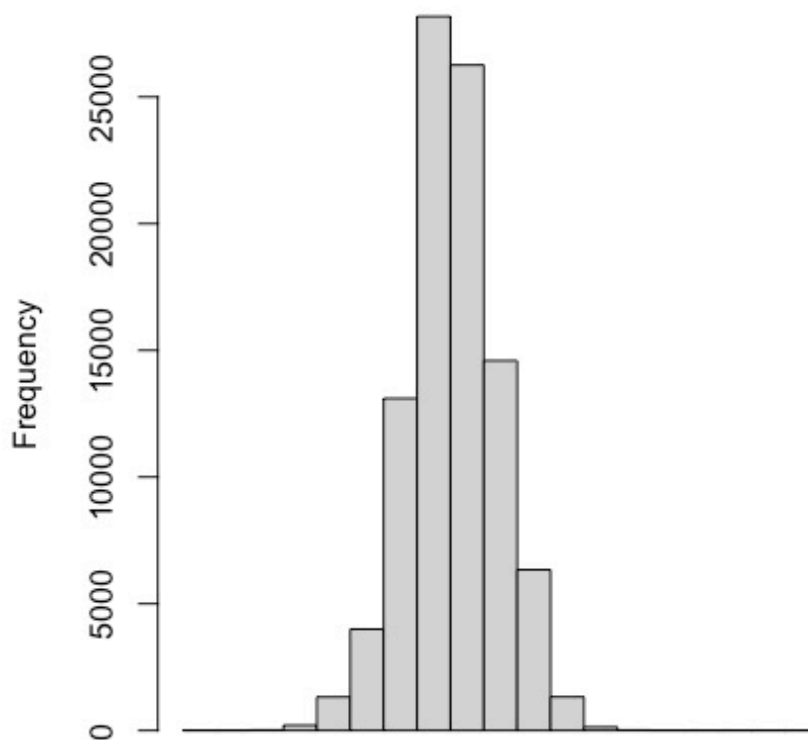
When calculated the correlation coefficient between the consumption expenditure and the social groups (caste specifically) was 0.172475. This is above 0 and is a positive value therefore there exists a positive relationship between the variables. This indicated that as progress from SC to 'other' castes, there is an increasing percentage change in consumption expenditure of the households. This supports our hypothesis.

- Homoscedasticity

When plotting the two variables in a histogram to check for homoscedasticity, we find that the spread of the distribution visualises in a bell-shaped curve. This indicates a homoscedastic spread which means that there is no increase in the variance of the error terms and their spread is constant.

Histogram of the residual values from Regc

Histogram of regc\$residuals

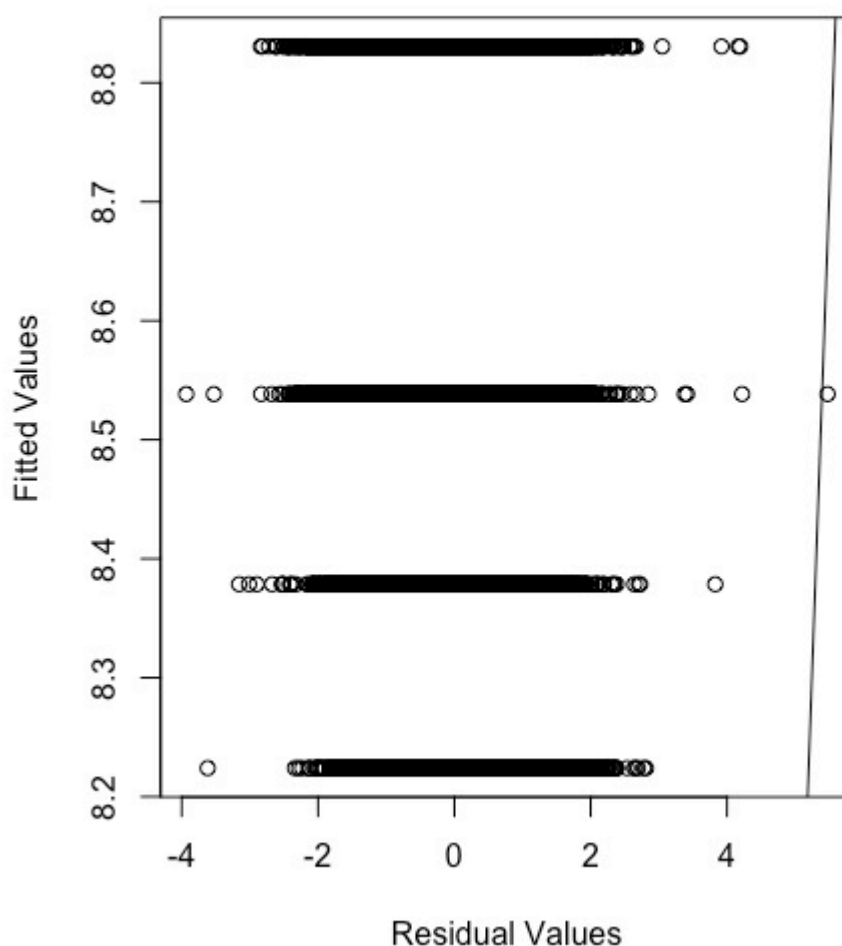


Graph 3

- No relation between residuals and fitted values

The residual values and the fitted values that were derived from this regression were plotted to understand if there was any relationship between them. There was a lack of uniformity in the scatter of the values, this indicates the absence of a relationship between the residual values and the fitted values. When fitting a qqline to the plot it was found to lie outside the cluster of the points and was lying to the extreme right of the plot, this further indicates an absence of any relationship. This further speaks to the accuracy of the regression we constructed.

Qqplot between residual and fitted values



Graph 4

REGRESSION ANALYSIS REPORT

Combined regression summary table for Regc, Rear, Rega and Reg

	Models			
	Dependent variable:			
	log(b3_q22)			
	Caste only (1)	Caste + Religion (2)	Caste, Religion + Age (3)	Caste, Religion, Age + Rural (4)
ST	-0.606*** (0.008)	-0.625*** (0.008)	-0.583*** (0.008)	-0.425*** (0.008)
SC	-0.452*** (0.006)	-0.463*** (0.006)	-0.452*** (0.006)	-0.352*** (0.006)
Other_backward	-0.292*** (0.005)	-0.280*** (0.005)	-0.274*** (0.005)	-0.205*** (0.005)
Age_below_18			-0.932*** (0.014)	-1.003*** (0.013)
Islam		-0.074*** (0.007)	-0.067*** (0.006)	-0.058*** (0.006)
Christian		0.304*** (0.014)	0.293*** (0.014)	0.232*** (0.013)
Sikh		0.489*** (0.017)	0.479*** (0.016)	0.522*** (0.015)
Jain		0.341*** (0.037)	0.332*** (0.036)	0.202*** (0.034)
Buddhist		0.177*** (0.027)	0.188*** (0.027)	0.081*** (0.025)
Zoroastrian		0.562*** (0.204)	0.550*** (0.200)	0.295 (0.188)
Other_religion		0.117*** (0.041)	0.084** (0.040)	0.076** (0.038)
Rural				-0.467*** (0.004)
Constant	8.831*** (0.004)	8.820*** (0.004)	8.832*** (0.004)	9.091*** (0.004)
Observations	95,511	95,511	95,511	95,511
R ²	0.083	0.098	0.138	0.235
Adjusted R ²	0.083	0.098	0.138	0.235
Residual Std. Error	33.483 (df = 95507)	33.203 (df = 95500)	32.460 (df = 95499)	30.571 (df = 95498)
F Statistic	2,875.578*** (df = 3; 95507)	1,039.735*** (df = 10; 95500)	1,390.789*** (df = 11; 95499)	2,451.303*** (df = 12; 95498)

Note: * p<0.1; ** p<0.05; *** p<0.01

Table 1

INTERPRETATION

Our aim was to see the relationship between social groups and their consumption expenditures. Through the above regression results, we can see how consumption expenditures changes in percentage terms as we consider more facets of an individual's social identity. We derive the effect of belonging to a certain social group by multiplying the coefficient of the chosen variable with 100 and equating that as the percentage change in their consumption expenditure.

$$\% \text{ change}(y) = 100 * \beta_1 * (x_1)$$

(This describes the percentage change in consumption expenditure with respect to the value of 100*(coefficient of the social group in consideration).

Coefficients and Intercept

1- Regc (ln(Consumption Expenditure) ~ Caste)

In this regression we determined the base characteristic of a household as belonging to the 'other' caste category and interpreted the coefficient of the other categories in accordance to that group.

We found that the households belonging to STs had the highest percentage decrease in consumption expenditure(60.6%) compared to 'Other' caste. On the other hand, OBCs had the lowest percentage decrease in consumption expenditure(29.2%) compared to 'Other' caste.

2- Regr (ln(Consumption Expenditure) ~ Caste+Religion)

In this regression we determined the base characteristic of a household as belonging to 'other' caste and to be a Hindu household. We interpreted the coefficients of the other categories in accordance to that group.

We found that STs still had the highest percentage decrease in consumption expenditure among the castes (62.5%) compared to the base household. In similar comparison, households who followed Islam(7.4%) as their religion had the highest percentage decrease in consumption expenditure among the religion groups. Zoroastrians contrarily have the highest percentage increase in consumption expenditure(56.2%) compared to the base household.

3- Rega (ln(Consumption Expenditure) ~ Caste+Religion+Age)

In this regression we determined the base characteristic of a household as belonging to 'other' caste, to be a Hindu household and for the head of the household to be above the age of 18. We interpreted the coefficients of the other categories in accordance to that group.

Similar trends were observed with respect to caste and religion as they were in Regr. Though it can be noticed that households whose head was below the age of 18 had a percentage decrease in consumption expenditure by 93.2% than households whose head was below the age of 18 and had the other base attributes of a household.

4- Reg (ln(Consumption Expenditure) ~ Caste+Religion+Age+Rural)

In this regression we determined the base characteristic of a household as belonging to 'other' caste, to be a Hindu household, for the head of the household to be above the age of 18 and for them to reside in an urban setting. We interpreted the coefficients of the other categories in accordance to that group.

Similar trends were observed with respect to caste and age of the household head as they were in Rega. An extreme case can be noted with respect to age where there was a decrease of 100.3% in consumption expenditure given the head of the household was below the age of 18.

With respect to other characteristics, it can be noticed that households who followed Sikhism have the highest percentage increase in consumption expenditure among the religion categories with Islamic households consistently having the highest percentage decrease in consumption expenditure across all the regression models. It was also found in this model that people residing in rural settings reflect a 46.7% decrease in their consumption expenditure than people who reside in urban settings and belonging to the base household.

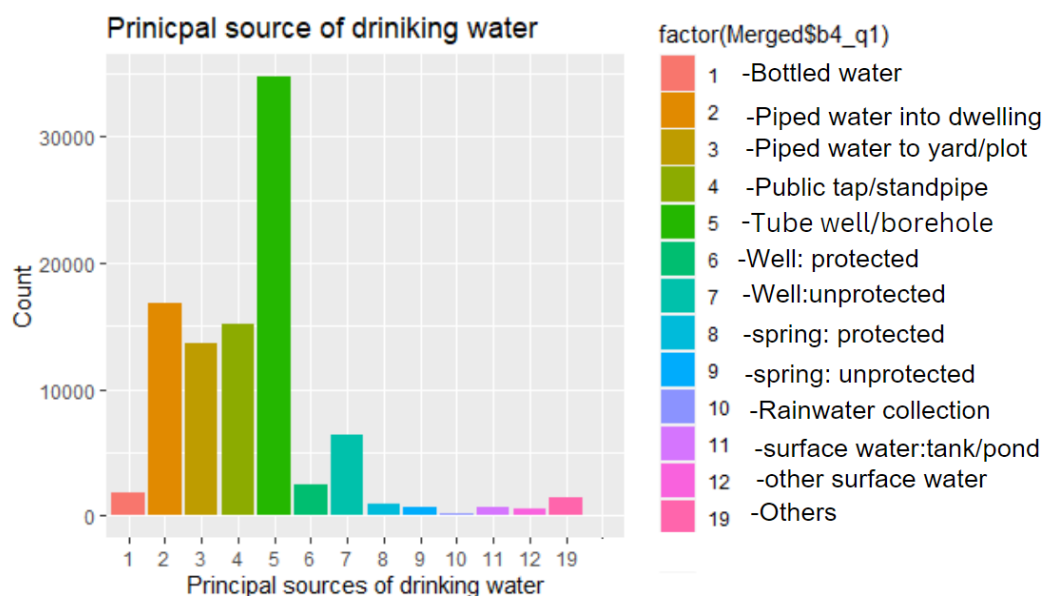
R-Squared Value

R squared is increasing across the models, this shows that there is an increasing correlation between the independent and dependent variables as we include more independent variables (social characteristics) in the regression. The r-squared increases from 0.083 in the Regc model to 0.235 in the Reg model. This indicates that the percentage change in consumption expenditure is being explained meaningfully by different compositions of caste, religion and the region(urban/rural) considered in the models.

P-Value

The p-value of this regression is less than 0.05 which means that the values we arrived to are significant and the regression reflects a meaningful correlation.

TIME TAKEN TO ACCESS PRINCIPAL SOURCES OF DRINKING WATER ~ SOCIAL GROUPS



Graph 4

METHODOLOGY

Initially, we have observed the principal source of drinking water of all households. After studying the data on drinking water, we saw that the highest used principal source of drinking water is tube/bore well. A question that immediately came to mind was whether these wells were privately owned and exclusively accessed or whether they were being used communally.

Question 6 in block 4, asks the question who fetches drinking water, a question which only takes answers when the distance from the principle source of water is greater than 0.2km from the dwelling. These to us implied some capturing of use of communal sources of water within the data set. We also had access to the time taken to reach the principle source and also how long they would have to wait to fetch water (questions 7 and 8 respectively).

Given India's history of heirarchially ordered division of labour and social priority our hope was that we could capture the extent to which it is at play with regard to accessing common sources of water, along lines of caste and gender. With respect to caste especially we expected to see some effect, given the practice of untouchability.

TESTING THE SUITABILITY OF THE REGRESSION

(I) Correlation Coefficient

When we calculated the correlation coefficient between the 'time taken to reach the source of drinking water and waiting time' and the social groups, it equated approximately to zero. Suggesting that there is no linear relationship between both these variables.

(li) Linearity

When we tried fitting a line through the scatter between 'time taken to reach the source of drinking water and waiting time' and the social groups, we observed a line but that's not saying their is a relationship between both these variables. It was merely a line which we forcefully fit in the scatter plot.

(lii) Homoscedasticity

When we tried observing the homoscedasticity, we have seen increase in the variation of errors and the regression is not homoscadastic. For this, we have plotted an histogram and it is right skewed. Suggesting that the spread of errors is not constant.



Graph 5

(Iv) Relation Between Residuals and Fitted Values

Further, we tried seeing the relationship between residuals and fitted values by plotting both the variables. The scatter plot showed that there is some uniformity between the variables, which is not supportive to draw a regression analysis.

These are the results when we ran the regression.

	models		
	<i>Dependent variable:</i>		
		b4_q7	
	OLS	Including caste variables	Including interaction terms
	(1)	(2)	(3)
SC		3.084*** (0.383)	3.331*** (0.391)
ST		5.829*** (0.433)	5.829*** (0.433)
Other_backward		4.656*** (0.344)	4.645*** (0.344)
Male_below_18	-0.189 (1.153)	37.998*** (7.848)	40.512*** (7.867)
Male_above_18	-3.447*** (0.662)	35.101*** (7.792)	35.057*** (7.790)
Female_below_18		38.009*** (7.806)	38.374*** (7.812)
Female_above_18	-3.608*** (0.590)	34.459*** (7.786)	34.416*** (7.784)
SC:Female_below_18			-1.522 (1.317)
SC:Male_below_18			-10.567*** (2.348)
non_mem_of_house	-38.178 (102.960)	3.476 (102.929)	3.486 (102.904)
Others	-38.178 (23.847)		
Constant	38.178*** (0.572)	-3.476 (7.788)	-3.486 (7.786)
Observations	40,906	40,906	40,906
R ²	0.001	0.008	0.008
Adjusted R ²	0.001	0.007	0.008
Residual Std. Error	1,314.488 (df = 40900)	1,310.336 (df = 40897)	1,310.027 (df = 40895)
F Statistic	9.979*** (df = 5; 40900)	39.105*** (df = 8; 40897)	33.427*** (df = 10; 40895)

Note:

*n≤0.1; **n≤0.05; ***n≤0.01

Our above checks show that given the nature of the data our planned result could not be run effectively. Running the regression perfectly encapsulates nonsense. Running an OLS model shows returns coefficients of -38.178 for both others and non-members of the household, while the intercept term is also, exactly 38.178. Given our coefficients are returning the amount of time it would take each individual above or below the intercept, it implies that both these categories are collecting the water in 0 minutes.

For our second model we shifted the base variable to others, such that the coefficients would return to us how long it would take for an individual who belongs to this mysterious others category, who also happens to be of general caste (such that we can study the difference in time taken between each of the marginalized castes in comparison to the experience of the general category). Now the intercept is negative, while previously those belonging to the others category took no time to fetch water, they now do it -3.4 minutes.

Beyond this, with each successive model the R-squared value of the model falls further, implying the fit is worse.

WORK DISTRIBUTION

REGRESSION 1 - Anjali

REGRESSION 2 - Maharshi